

Homework 1

For practicing – no grading

ECE 253 - Digital Image Processing - Fall 2025

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1. Let $h(n)$ be the impulse response of an LTI system:

$$h(n) = \delta(n) - 2\delta(n - 1)$$

- (a) Compute the DTFT $H(e^{j\omega})$
 - (b) Compute the z-transform $H(z)$
 - (c) Plot the pole-zero on the unit circle
 - (d) Compute and sketch $|H(e^{j\omega})|$
 - (e) Compute and sketch $\angle H(e^{j\omega})$
 - (f) Find the output $y(n)$ for $x(n) = \delta(n) + \delta(n - 1) + \delta(n - 2)$
2. Given $h(n)$ in problem 1 above, let $h_1(n) = (-1)^n h(n)$
- (a) Find and sketch $h_1(n)$
 - (b) What is the relationship between $H_1(z)$ and $H(z)$; $H_1(e^{j\omega})$ and $H(e^{j\omega})$?
 - (c) Plot the pole-zero of $H_1(z)$ on the unit circle
 - (d) Compute and sketch: $|H_1(e^{j\omega})|$ and $\angle H_1(e^{j\omega})$
3. $G_1(z) = 1 + \frac{1}{3}z^{-1}$
- (a) Find $G_2(z)$ such that $G(z) = G_1(z)G_2(z)$ is a linear-phase system.
 - (b) Find $G(z)$. Compute its magnitude and phase responses.
 - (c) Plot the pole-zero of $G(z)$ on the unit circle
 - (d) Which linear-phase type is $G(z)$?
4. Given $G(z)$ in problem 3, add zeros to make a Type II linear-phase system $G_{II}(z)$.
- (a) Plot the pole-zero of $G_{II}(z)$
 - (b) Find $G_{II}(z)$
 - (c) Compute and sketch $|G_{II}(e^{j\omega})|$ and $\angle G_{II}(e^{j\omega})$

5. Let $h_2(n) = h(n) * h_1(n)$, where $h(n)$ and $h_1(n)$ are from the previous problems.
- (a) Compute the convolution $h_2(n)$ in the time domain.
 - (b) Compute the z-transform $H_2(z)$ and find the pole-zero plot.
 - (c) Compute the DTFT $H_2(e^{j\omega})$ and sketch the magnitude and phase responses.
 - (d) Determine whether $h_2(n)$ is causal and stable. Why?
6. Consider a system with an input $x(n) = \cos(\omega_0 n)$ and impulse response $h(n)$ from Problem 1.
- (a) Compute the output $y(n)$ using the frequency response $H(e^{j\omega})$ of the system.
 - (b) If $\omega_0 = \frac{\pi}{3}$, compute the exact output signal $y(n)$.
 - (c) Discuss whether the system acts as a low-pass, high-pass, band-pass, or band-stop filter based on the magnitude response of $H(e^{j\omega})$.
 - (d) Sketch the output signal $y(n)$ for the first 10 samples when $\omega_0 = \frac{\pi}{3}$.